

Diligently Create The Cradle Of Life

Infant Phototherapy Unit

AL-10D

(V2.5)

User manual

READ AND KEEP THIS MANUAL CAREFULLY PLEASE

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Contents

1	Summary	2
2	Use	2
3	Symbols and marks	2
4	Features	2
5	Main technical parameter:	3
6	Installation	3
7	Use of the device	4
8	Panel function introduction.....	3
9	Precautions	4
10	Warnings	4
11	Maintenance and troubleshooting	5
12	After-sale service	6
13	Encasement inventory	7

AL-10D infant phototherapy unit

1 Summary

Jaundice is a common phenomenon in neonatal period, it is observed that over 90% newborn babies have jaundice in different degree. It can be a physiological phenomenon (called physiological jaundice) in neonatal period, it also can be an important symptom of many diseases in neonatal period (called pathological jaundice) . This pathological jaundice can cause bilirubin brain illness (called kernicterus), it can cause brain damage of newborn baby, and results in early death and serious sequela, so we must take immediate and efficient measures to treat neonatal pathological jaundice.

The effect of phototherapy treatment for neonatal high bilirubin blood illness has been approved by clinic, its principle is that bilirubin can absorb rays to have actinic isomerization, it makes the indirect bilirubin oxidize into one water-soluble product under the bluish green light (light-oxidation bilirubin).It fades yellow according to liver and gallbladder discharging out of body. The medical field recognizes that using phototherapy to treat neonatal pathological jaundice is a simple and effective method. Neonatal phototherapy unit are also called "infant phototherapy device" or "blue light lamp". It is necessary medical equipment of neonatal medical department.

2 Use

This product is used for light treatment of high bilirubin for premature baby or newborn baby.

3 Symbols and marks

== 12V 1.2A

LED light box power input: DC 12V 1.2A



Attention, consult accompanying documents



When using phototherapy, eye shields for the patient shall be used.

4 Features

- 4.1 Electrical adapter, the voltage range is very wide.
- 4.2 LED radiance, high efficiency, long life-span.
- 4.3 The size is small and weight is light

5 Main technical parameter:

- 5.1 Adapter power supply input: AC 110V/60Hz; adapter output: DC 12V 1.2A
- 5.2 LED light box power input: 35VA×2
- 5.3 Wave length range: 420—490nm
- 5.4 Total Irradiance for bilirubin: 1500 μ w/cm²
- 5.5 Environment temperature: +18°C—+30°C
- 5.6 Noise level: $\leq$ 55dB(A)
- 5.7 Working condition:
 - a. Ambient temperature: +18°C~+30°C
 - b. Relative humidity range: 10%~85%
 - c. Atmospheric pressure: 700hpa~1060hpa
- 5.8 Transport and store condition:
 - a. Environment temperature range: -10°C—+55°C
 - b. Relative humidity range: $\leq$ 80%
 - c. Atmospheric pressure: 500hpa~1060hpa

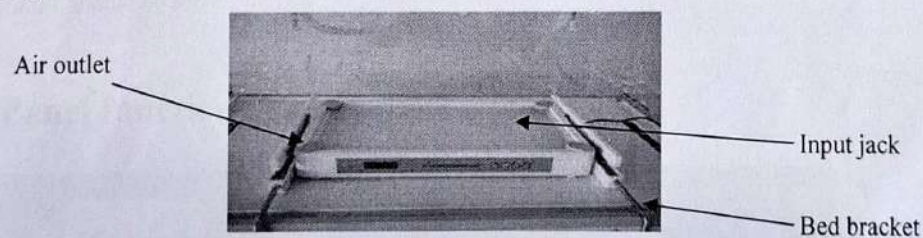
6 Installation

Instruction: This type phototherapy unit can be installed in the baby incubators. It is used under Perspex infant bed. So before installing this phototherapy unit, please install the infant incubator first, then install this phototherapy unit.

The installing procedure of this phototherapy unit is as follows:

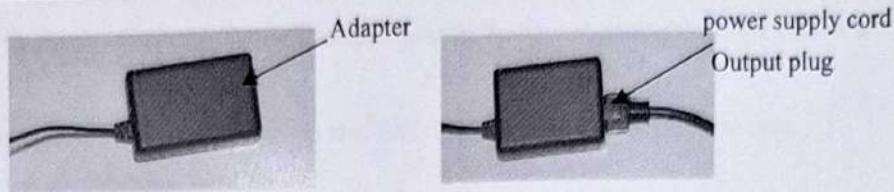
- 6.1 Pull out the power supply plug of the infant incubator to make the device power off.
- 6.2 Open the front door of the hood, take out Perspex infant bed.
- 6.3 Put this device in the middle of the infant bed shelf.

Notice: Put the Perspex side of the device upward and DC input jack is in the back. (see picture 1)



Picture 1. Putting the phototherapy unit

- 6.4 Insert the output wire (DC 12V) into the power supply input jack through infusion hole.
- 6.5 Put on Perspex infant bed, close the front door of the hood, and lock it tightly.
- 6.6 Insert one end of the power supply cord into the adapter. (see picture 2)



Picture 2. Connecting adapter input wire

7 Use of the device

7.1 You must clean this device before using, especially that there mustn't be dirt on the surface of the light box.

7.2 Insert the power plug of the baby incubator into the hospital's AC 110V/50Hz power single-phase three-hole socket (note that a single-phase three-hole socket with a good ground must be used).

7.3. Make the incubator work in the temperature control mode and set it to 33°C for 1 hour.

7.4 Limit the baby to the effective surface of the crib (the center of the effective surface is located in the center of the crib), and implement protective operations such as eye masks and perineal masks for the children.

7.5. After wearing eye masks and perineal masks to children, implement light-blocking protection around the equipment.

7.6 Press the blue light switch, the blue light can glow, the timer on the panel starts timing, the phototherapy starts, and the digital tube directly displays the time of this phototherapy. The maximum time of this phototherapy is 100 hours (or turn off the switch) after phototherapy and Timing ends.

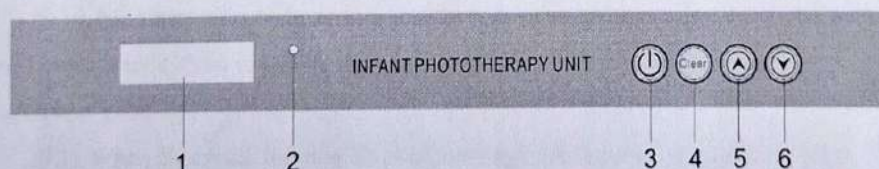
Note: Children's body temperature should be checked regularly during phototherapy.

And according to the situation to give the baby appropriate supplement liquid.

Note: Intermittent radiation therapy can be implemented when necessary.

After the phototherapy is over, the control mode of the incubator should be reselected and the temperature should be set.

8 Panel function introduction



1 **Time display window**---display each treatment time and total cumulative time.

2 **Over-temperature indicator light**---when the temperature in the phototherapy lamp box exceeds a predetermined value, the red indicator light will light up.

3 **"Start/Stop" button**---press once to start irradiation, press again to stop irradiation.

- 4 **"Clear" button**----press this button for a few seconds, the current time of the timer is reset to zero.
- 5 **"Increase" key**——After pressing this key, the current timing display returns to zero. Then you can set this irradiation time.
- 6 **"Decrease" key**-----press this key to display the cumulative time of the timer.

Figure 8-1 The front panel of the blue light therapy device

9 Precautions

9.1 When using, you must lock the brake castors of baby incubator tightly for fear of that the equipment moves.

9.2 You should clean the light box and the surface of the Perspex infant bed frequently to keep them clean and transparent.

9.3 When phototherapy, the patient's radiated skin should be nude, and you should check the bed temperature and radiated skin situation on time (no more than 4 hours). and the patient's gesture should be changed properly. For those patients with weak movements, the doctor can implement intermittent radiance to reduce side effect.

9.4 Using phototherapy unit will influence the body temperature of the patient, it is very necessary to measure the baby's temperature. It also may affect some patients' water balance. So the doctors should supply liquid for the baby for fear of lacking of water.

9.5 When phototherapy, the doctor should measure the patient's bilirubin on time.

9.6 You should check the fan of the phototherapy and the surface temperature of the infant bed on time. If you find something abnormal, you should stop using the product immediately.

9.7 To guarantee the clinical phototherapy effect, the LED lamps should be changed when using about 20000 hours.

9.8 You must pay attention to the changes of the environment (for example, temperature, different radiant source, etc), these can also affect the treatment effect.

9.9 When phototherapy, we advise that the incubator should use skin temperature control mode. Those incubators that only has air temperature control mode or are set to air temperature control mode, the setting temperature must be reduced according to the baby's body temperature.

9.10 For those who have serious jaundice, besides phototherapy treatment, they also need medicine to trade, even use changing blood treatment for fear of only use phototherapy, the effect is not very good.

9.11 When observing the skin color of the baby, you need to use the white light, because the blue light can affect observing skin.

9.12 When repairing, you can only use our appointed parts (including adapters and LED lamps), otherwise it will reduce the safety and validity of the device. This product's life span is 6 years, at that time, you should scrap the product according to relevant laws or send it to our company.

10 Warnings

10.1 During phototherapy, the doctor should use black cloth or eye masks to cover the baby's eye, because the blue light is harmful for patient's retina, and the eye mask should be suitable.

10.2 During phototherapy, the patient's genitalia should be covered by diaper, for fear of damaging the function of genitalia.

10.3 During phototherapy, the isomer of bilirubin may cause toxic effect.

10.4 The patient who is near phototherapy unit can use lighttight black cloth for protection, if the operator is under the blue light for long time, he may also be affected.

10.5 Using lighttight cloth to cover the hood for protection may affect the nursing and observation of the patient, so we advise that the nurse should observe the baby frequently, if necessary, you should reserve the transparent observing window to observe.

10.6 Because of the actinism of the drugs, storing the drugs and infusion fluid in the blue light area is forbidden.

10.7 Using flammable preparation (antiseptic, detergent) for the therapy unit is forbidden.

10.8 Using organic solvent such as alcohol, etc, to clean the surface of the Perspex is forbidden, covering radiator hole of the light box is forbidden.

10.9 Using the therapy unit to irradiate the patient with low bilirubin is forbidden.

11 Maintenance and troubleshooting

11.1 When not using the device, you should pull out the power supply plug, and put the adapter well.

11.2 Use clean cloth to clean the device, especially that there should not be dirt on the surface of the light box to keep it clean and bright. (we advise that you use soft cotton yarn dipping with water liquor of neutral detergent to clean the surface dirt of the Perspex, then dip with distilled water to wash, finally, use soft and dry cloth to clean it).

11.3 The radiance of the LED lamp will reduce as time passes, you should change for new LED lamps in time to guarantee the phototherapy effect.

11.4 Common trouble (see table 1)

Table 1:

Problem	Cause analyze	Solutions
Insert the power supply plug, all the LED lamps are not lightening	1. Power supply cord not insert well 2. 12V plug not insert well 3. The adapter is broken	Insert the power supply cord well Insert the plug well Change the adapter
Insert the power supply plug, some LED lamps are not lightening	1. LED lamps are broken 2. The restrict current resistances are broken.	Change for new LED lamps Change for new restrict current resistances.

12 Encasement inventory

Please check the components on the list when you open the case..

SN	Name	Quantity	Fully Note
1	Phototherapy light box	1	
2	Light box adapter	1	
3	Power supply cord	1	
4	Operation Manual	1	V2.5
5	Certificate of quality	1	